

# Muthu Ramaswamy

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## EDUCATION

### Singapore University of Technology and Design

- Bachelor of Engineering (Computer Science and Design), Honours with Distinction
- Minor in Artificial Intelligence
- Date of graduation: May 2026

Singapore

Jul 22 to May 26

## WORK EXPERIENCE

### KLASS Engineering and Solutions

May 25 to Sep 25

#### Machine Learning Research Engineer Intern, Spring Term

- Benchmarked multi-stage license plate recognition pipelines (3-stages of YOLOv7 vs 2-stage YOLOv10 + PARSeq) across datasets, analyzing results and documenting findings, identifying the 2-stage pipeline as superior.
- Fine-tuned a pretrained all-night license plate character detection model (PARSeq) on day/night datasets, improving daytime recognition accuracy while maintaining nighttime performance.
- Automated dataset curation, including file transfers, XML annotation filtering, and element mapping using Python scripts.
- Developed Streamlit interfaces to streamline human-in-the-loop annotation and review processes.
- Gained hands-on experience with Linux command line tools while managing data pipelines and model training tasks.

### SATS Ltd

Sep 24 to Dec 24

#### Data Analyst Intern, Fall Term

- Trained a YOLOv8 model for object detection using Python as part of a proof-of-concept project, focusing on warehouse shipments and QR codes, optimizing performance through hyperparameter tuning.
- Devised a PTZ camera control algorithm to enhance computer vision-based detection, ensuring efficient preset transitions and accurate recognition in dynamic environments.
- Automated data preparation by converting COCO annotations to YOLO format and created scripts for video processing, frame capture, and blob storage integration.
- Adapted quickly to new technologies such as Azure Machine Learning and Blob Storage to streamline data management and model experimentation.

## ACADEMIC PROJECTS

### Personal Project

Jun 26 to Jul 26

#### DataPilot - Agentic AI-Powered Data Science Assistant

- Developed an agentic AI data assistant using LangChain, LangGraph, and large language model tool calling to automate dataset exploration, profiling, cleaning, visualization, and manipulation through natural language.
- Built custom tools for automated dataset profiling, including directory inspection, split analysis, annotation format detection, and image-label consistency validation.
- Implemented sandboxed Python execution with human-in-the-loop approval, enabling the agent to safely generate and execute analysis and data-processing scripts on an isolated copy of the dataset while preserving the original data.

### Singapore University of Technology and Design

Feb 25 to Apr 25

#### Spoken Language Identification Deep Learning Project

- Created a spoken language identification system on the Mozilla Common Voice dataset (9 languages) in a team of 3.
- Designed CNN-BiLSTM-Attention architectures (ResNet34/50 + self-attention) to capture acoustic and temporal features.
- Built a modular PyTorch Trainer supporting configurable training loops, early stopping and performance visualization.
- Implemented an audio preprocessing pipeline: MP3 to WAV conversion, 16kHz resampling, speech activity detection and log-mel spectrogram extraction.

### Singapore University of Technology and Design

May 24 to Aug 24

#### Dell Project: Resource Management App

- Collaborated in a team of 7 to develop and deploy a comprehensive web application for workshop resource management.
- Implemented RESTful APIs using Node.js and Express.js to handle requests and resource management.
- Utilised MongoDB as the database to ensure efficient data storage, retrieval and manipulation.
- Designed and executed unit/integration tests (Jest) and end-to-end system tests (Cypress), strengthening system reliability.

## ADDITIONAL INFORMATION

- Programming/Technical Skills: Python (PyTorch, Streamlit, Flask, Pandas, Scikit-learn), JavaScript (Node.js, Express.js, React), SQL, MongoDB, R, HTML/CSS, Java, Go, LangChain for Agentic AI
- Tools: Visual Studio Code, Docker, Git, Linux, Azure ML, AWS, Tableau, Power BI, Android Studio, Adobe Illustrator, Excel
- Fluent in English and Tamil (spoken & written)